



In-circuit testing

internationally acceptable requirements, with IPC certifications available in India. The trainings start from the design, SMT process set-up, process validation, assembly, repair, rework and inspection. Innovative ESD training programmes, design for manufacturing, design for assembly and design for testability are some forthcoming areas. Programme management for CEMs is also an area of development that is expected to come up in India.

However, most of the trainings are customer-driven, specifically for contrast electronics manufacturing where the customer requirement pushes the training type. This is unlike OEMs, which have their manufacturing set-up in India and impart internal training as a new process requirement comes up.

Product data management

An electronic product involves multiple manufacturers in the supply chain at system or sub-system level—PCB fabrication, ASIC development and fabrication, PCB assembly, mechanical assembly and finally system integration and software installation.

Many a times these manufacturers refer to the product information specific to their deliverables. For example, the mechanical parts manufacturer who never gets to see PCB fabrication data like Geber (file format used by PCB industry software) may lack

valuable information which could be critical to the end-product's quality.

Many services are coming up from Indian companies that are increasingly participating in product data management. These services help to establish communication between the participating partners as well as efficiently change the management. The IPC-2581 standard developed by IPC and International Electronics Manufacturing Initiative (iNEMI) is an XML-based scheme representing intelligent data file format, which is increasingly being looked into by OEMs based out of India. These OEMs also participate in the IPC-2581 Consortium helping to improve the standard from Indian manufacturing's point of view.

Specialised services

With increasing PCB assembly complexity, specialised services are being provided by companies to OEMs and CEMs. These include design for eXcellence (design for reliability, design for testability and design for Six Sigma) and product data management services.

Reliability study services are also provided by companies. These services test various parameters according to product requirements—like highly accelerated life tests for failures-in-time calculations and high-temperature operating life tests which help failure detection at different times of the product life (like at 100, 500 and 1000 hours). Other tests include solder heat-resistance test, electromagnetic interference/electro-

magnetic compatibility tests and ESD-qualification tests as per IEC standards.

Third-party certification services in line with these testing services are showing a huge growth in India for products in various applications including telecommunication, military and healthcare. These tests and qualifications are mainly performed at the system level, which verifies the mechanical strength and various other parameters of SMD solder joints.

Specialised rework services for boards with surface-mount components is an upcoming trend in different stages of the board's life cycle—prototype, production and field returns. These include flip-chip components, package-on-package, inner-layer PCB laminate rework as well as rework of embedded discrete SMD components in the PCB.

Consultants for manufacturing process defects troubleshooting are sought by the Indian CEMs. Design consultation for new SMD component packages, thermal management and power rail design for multiple power rails on a board is also being sought by the Indian OEMs. Also, consultation for Six Sigma method and Japanese manufacturing methods like fool-proofing—in the form of Poka-Yoke and Kaizen (never-ending improvement process)—is being encouraged by the National Productivity Council. Clusters of Indian CEMs are being formed. Benchmarks within each cluster help improve manufacturing practices.

The way ahead

Use of SMT in electronics manufacturing is evolving. New services supporting SMT are helping to set new benchmarks and improve manufacturing practices. The lacunae identified within this segment will be filled over the years driven by other sub-segments. ●

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